LA-2 PROJECT

Smart Water Monitoring System

Environmental Science and Sustainability (CIV104)

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A simple idea to help notice unnecessary water flow.

1. Introduction

Water is important in our daily lives, so we should avoid wasting it. Sometimes water keeps flowing because a tap is left open or a plumbing problem is not noticed quickly. For this LA-2 project, we suggest a simple system that measures water flow and gives a warning when the flow seems unusually high.

The NSSKC visit handout mentions plumbing parts such as taps, pipes, valves and fittings. These can help us think about where water is used and where monitoring might be useful. Our group should add any specific lab observation only if we actually noticed it during the visit.

2. Aim

To show how a flow sensor, Arduino and a simple Python program can help users notice unusually high water flow.

3. How the System Works

1. A water-flow sensor measures the movement of water.
2. An Arduino reads the sensor signal and sends the reading to a computer.
3. A Python program displays the flow reading.
4. If the reading stays higher than the chosen limit, the program shows a warning so someone can check the tap or pipe.

4. Materials Required

- Water-flow sensor (for example, a YF-S201-type sensor)
- Arduino Uno
- USB cable and connecting wires
- Computer with Python installed

5. Example Warning Levels

These numbers are only sample values for a demonstration. The group should test and adjust them for the sensor and setup before using them.

Flow reading	Display
Up to 6 L/min	Normal
Above 6 to 10 L/min	Warning
Above 10 L/min	High flow — check if expected

6. Benefits

- Makes unusual water flow easier to notice.
- May help people respond to a possible plumbing issue sooner.
- Connects a practical water-saving idea with basic programming.

- Supports awareness of SDG 6: Clean Water and Sanitation, and SDG 11: Sustainable Cities and Communities.

7. Limitations

The system cannot repair a leak by itself, and a high reading does not always mean there is a leak. The sensor also needs to be tested so that its readings are reasonably accurate.

8. Team Work

The following are suggested roles. Change them if your group divided the work differently.

Member	task
Rahul (Admin & Lead Developer)	System Administrator & Lead Developer of this website; report QA
Sushanth	Coordinate the group
Rohan	Help connect the sensor and Arduino
Jilani	Check the sensor readings
Adthiya	Run the Python program
Soyug	Collect notes and help with the presentation

9. Conclusion

The Smart Water Monitoring System is a simple idea for reducing unnoticed water wastage. A sensor and Arduino can measure flow, while a Python program can show a warning when flow stays unusually high. The system would need to be tested before we can say how well it works, but it shows how technology can support careful water use.

References

1. LA-2 Handout, Environmental Science and Sustainability (CIV104), provided for the assignment.

2. NSSKC Visit Handout, provided for the assignment.